Customer Shopping Behavior Analysis

End-to-end retail data analytics — Python · SQL · PostgreSQL · Power BI

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Made with GAMMA

Project Objectives

01

Clean & Preprocess

Python & Pandas

02

Store Data

PostgreSQL

03

Query Insights

SQL analysis

04

Visualize

Power BI dashboard

05

Identify Trends

Categories, gender, seasonality

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Tools & Technologies

Python

Pandas, NumPy

Jupyter

Notebook

PostgreSQL

Database

SQL

Queries

Power BI

Dashboard

CSV

Dataset

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Dataset at a Glance

3,900

Total Records

18

Columns

Columns spanned customer demographics, product categories, purchase amounts, payment methods, review ratings, and subscription status.

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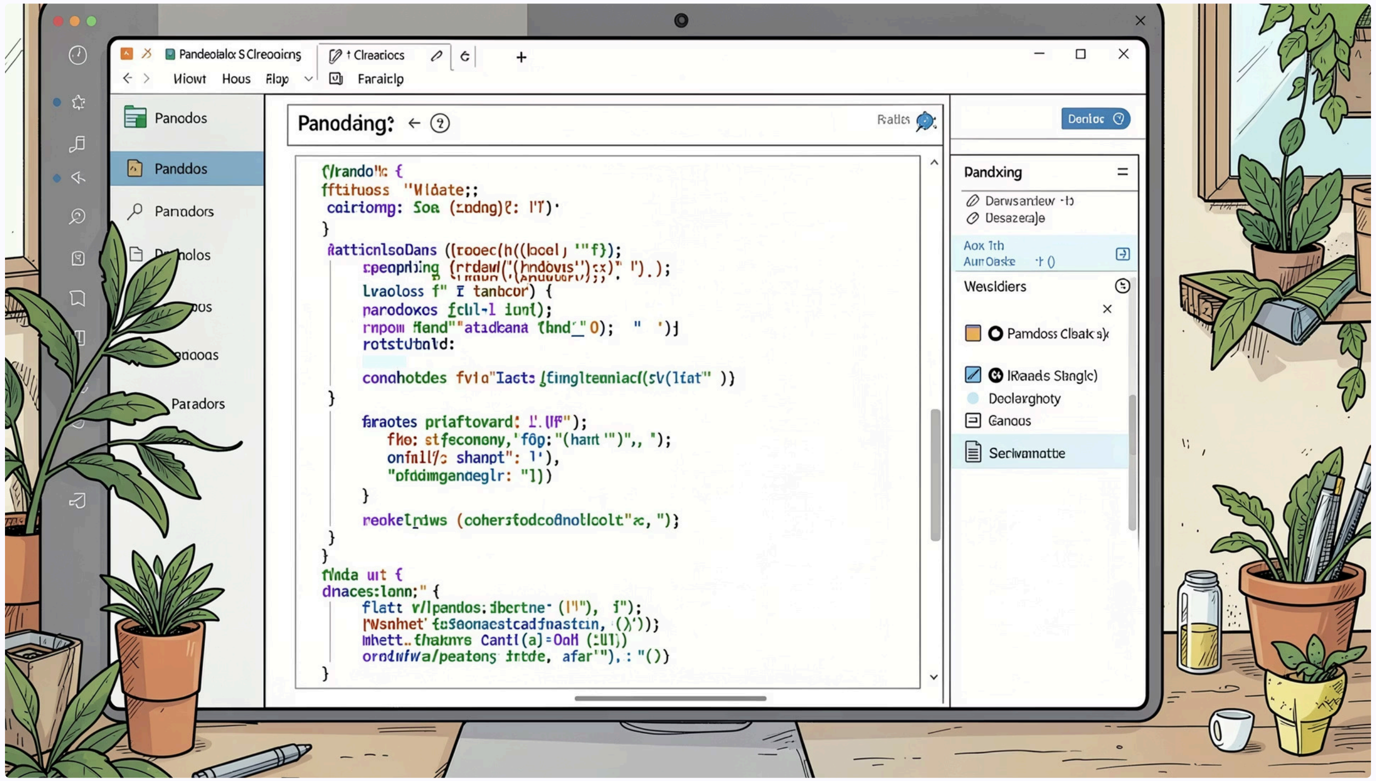
Data Cleaning & Preprocessing



Import dataset

Pandas

Handle missing values

Null entries

Create analytical columns

Categorization

Correct datatypes

Formatting

Prepare for integration

SQL & Power BI

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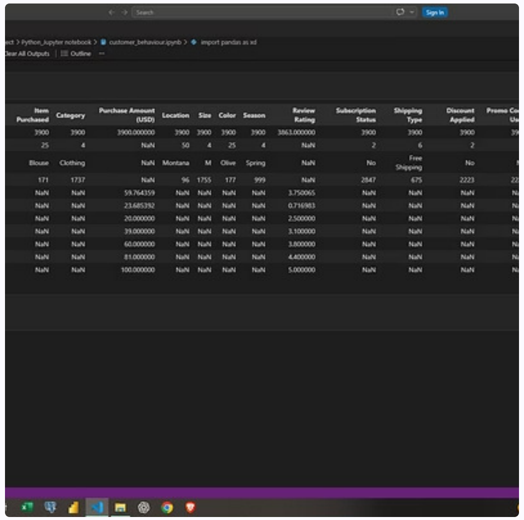
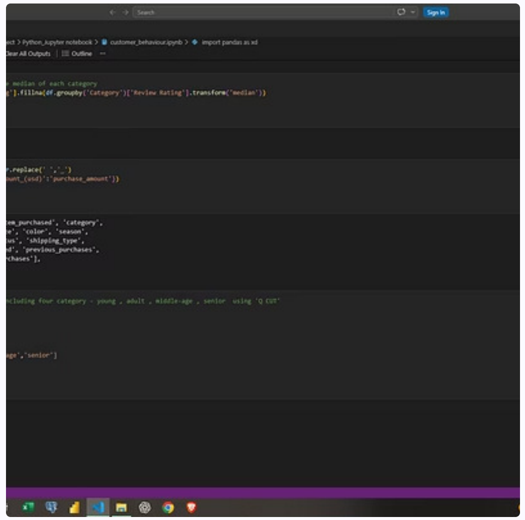
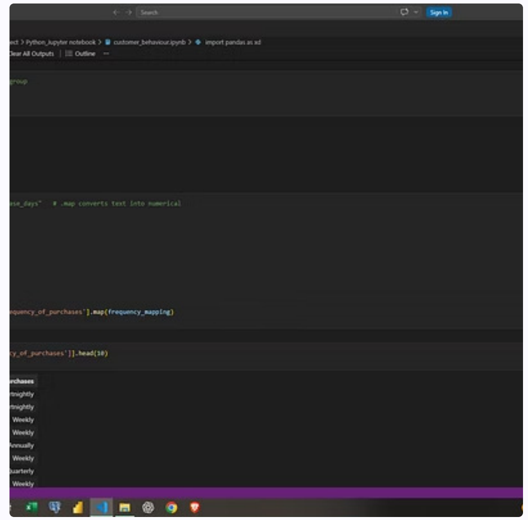
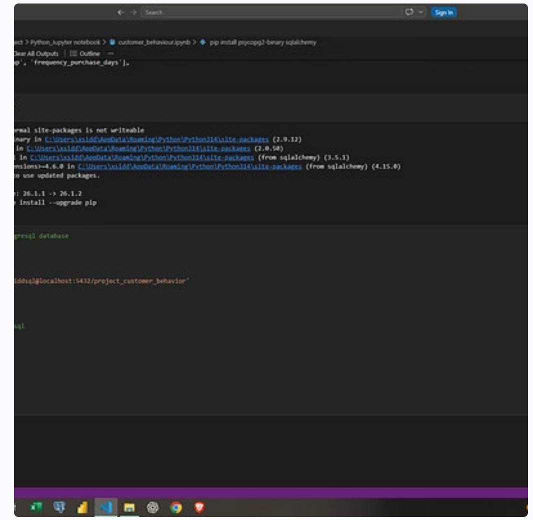
Python Data Cleaning

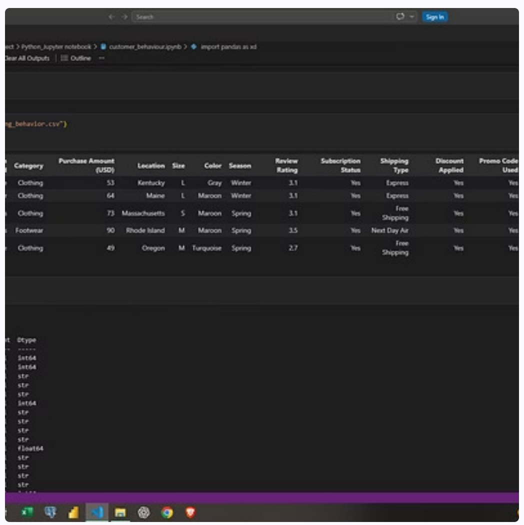
Step-by-step cleaning workflow in Jupyter Notebook using Pandas.











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SQL Analysis

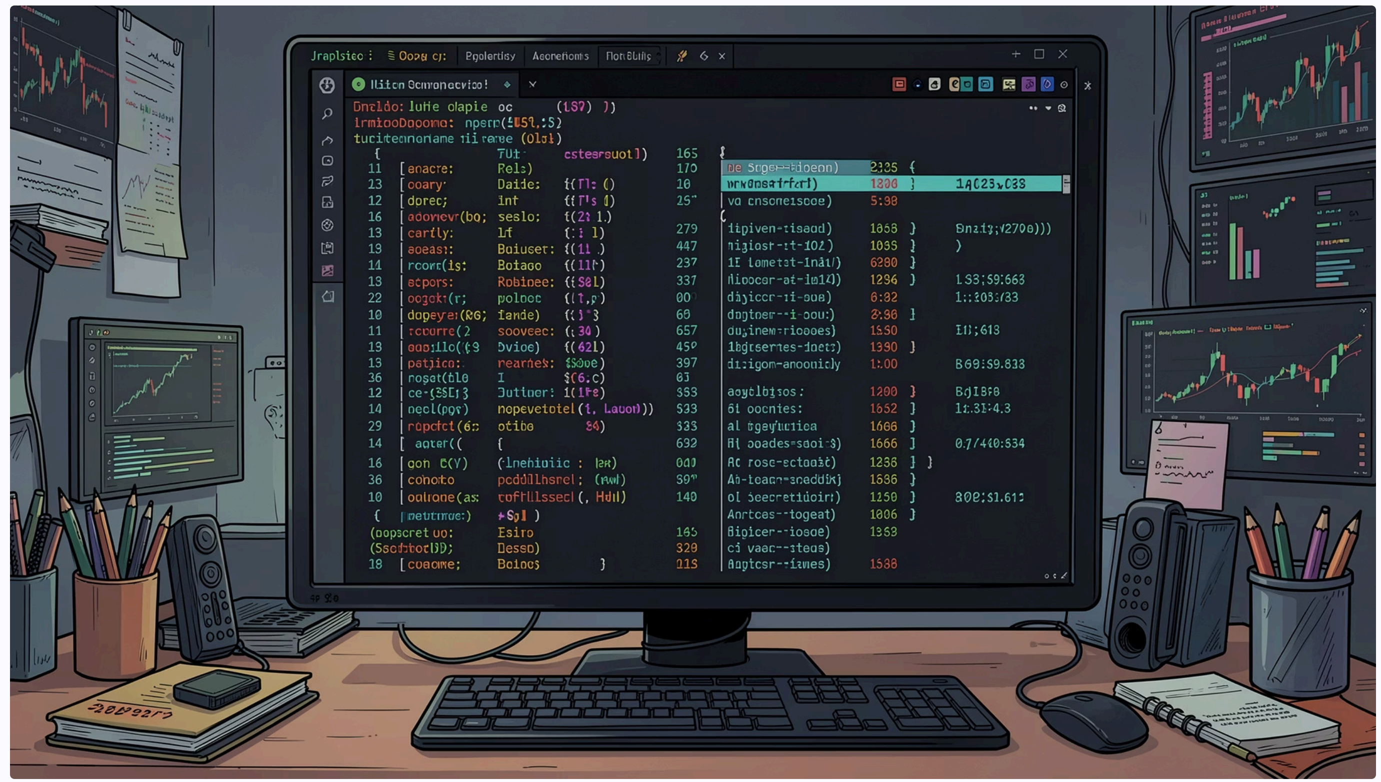
Revenue by Gender & Category

Top-Performing Categories

Seasonal Sales Trends

Subscription Behavior

Customer Purchasing Insights



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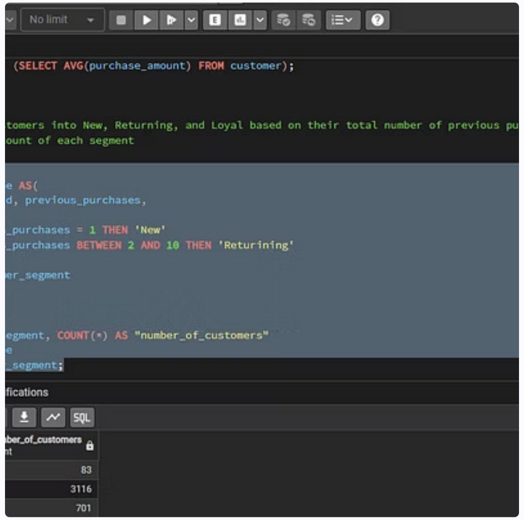
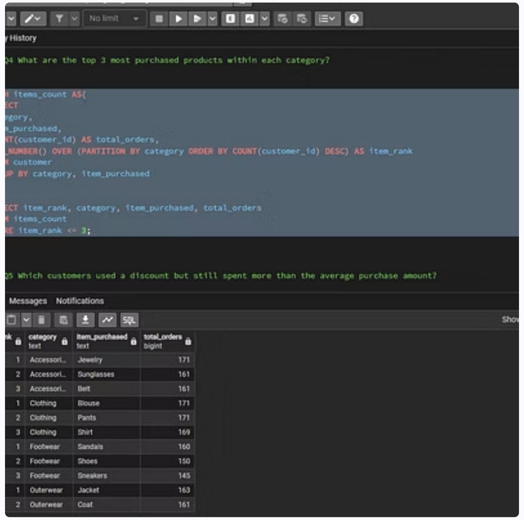
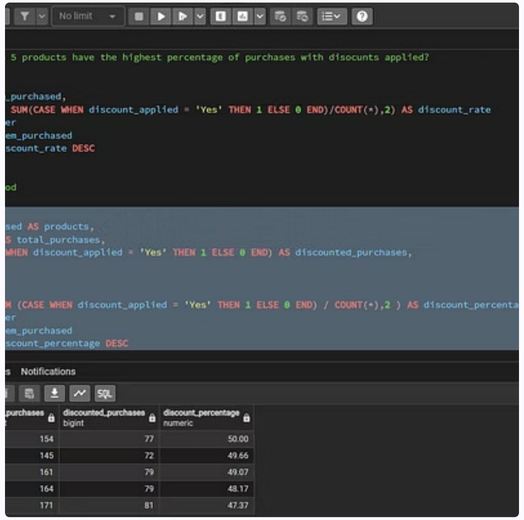
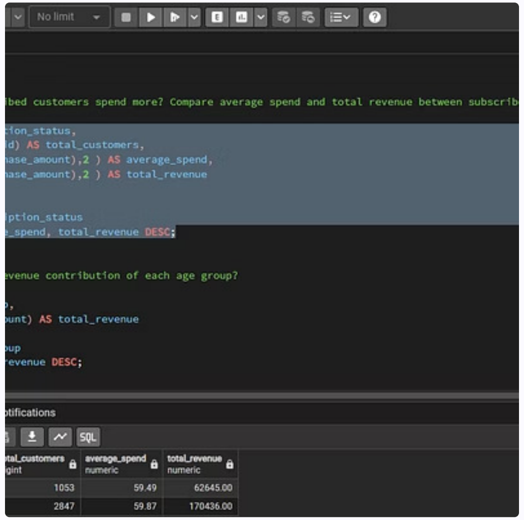
SQL Query Results









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Power BI Dashboard

Dashboard Components

KPI Cards

Seasonal Revenue Charts

Customer Demographics

Payment Methods

Subscription Analysis

Category Performance

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Key Insights & Conclusion

👕 Clothing

Highest revenue category

♂️ Male Customers

Larger revenue share

📅 Seasonality

Strong seasonal patterns

🔁 Subscriptions

Distinct purchasing behavior

💳 Payments

Nearly equal preference

End-to-end workflow: raw data → cleaned dataset → SQL insights → Power BI dashboard. Hands-on experience with Python, PostgreSQL, SQL, and Power BI.

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